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1. [5] What is the absolute (positive) difference between the mean and mode of the numbers 16, 32, 21, 16, and 20?
2. [5] **Sneha Kundu** and **Angie Huang** are moving around a circular track $\frac{1}{4}$ of a mile long. Because Sneha despises running, she walks at 2 miles per hour, and Angie runs at 12 miles per hour. How many laps around the track will Angie have completed once Sneha has walked one mile?
3. [5] Let $ABCDEF$ be a regular hexagon with side length 8. Find the area of quadrilateral $ABDF$.

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4. [6] **Aarya Garimella** is selling slime. He has two products: The *Slime Apprentice* kit, containing 14 pieces of slime and a cleanup kit, and the *Slime Master* kit, which contains 34 pieces of slime and a cleanup kit. If the Slime Apprentice kit costs 52 dollars to produce and the Slime Master kit costs 112 dollars to produce, what is the production cost of a single piece of slime, in dollars?
5. [6] Consider a circle inscribed in an ellipse with major axis of length 6 and minor axis of length 4. Compute the area of the region outside the circle but inside the ellipse.
6. [6] Positive integers a and b satisfy $ab = 18$. Compute the smallest possible value of $a^2 + b^2$.

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7. [7] The graph of $f(x) = x^2 - ax + b$ is symmetric about the line $x = 2$. Find a .
8. [7] Aaron is going for a run. He starts from his house and runs 6 miles east, 15 miles north, 14 miles west, then runs back home in a straight line. How many miles did he run in total?
9. [7] Right triangle $\triangle ABC$ is inscribed in a circle with diameter $BC = 5$. Let D be the foot of the altitude from A , and let AD intersect the circle at $E \neq A$. Given that $AB = 3$, find $5AE$.

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10. [9] For two-digit numbers with non-zero digits, a flipped version of a number is the number when its digits' order is reversed. For example, the flipped version of 12 is 21. If the sum of a number and its flipped version is divisible by some number n for all two-digit numbers with non-zero digits, find n .
11. [9] Compute $\left\lfloor \sqrt{20 - 2\sqrt{51}} \right\rfloor$. Here, $\lfloor x \rfloor$ denotes the largest integer not exceeding x .
12. [9] **Sneha Kundu** is running at her school's Fun Run (We cannot say which school as it would spoil a future problem). Unfortunately, she despises running and takes a break after each lap with probability $1/2$. Suppose Sneha's gym teacher has a $1/4$ chance of seeing her each time she takes a break. Compute the expected number of laps Sneha will run before getting caught.

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13. [11] A 5 foot tall horse has one end of a 5 foot long rope attached to the top of its head. The other end of the rope is attached to the top of an 8 foot tall pole of negligible thickness. If the horse is always upright, compute the area of the region where the horse can roam.
14. [11] Daren and Aaron live 16 miles apart. They want to meet up somewhere in their city, however, their moms insist that they cannot travel too far. As a result, Aaron and Daren's combined displacements (distances) from their respective houses must be less than 34 miles. Compute the area of the possible points where they could meet up in the city, in square miles.
15. [11] **Angie Huang, Austin Wang, Michael Luo, and Ryan Niu** are standing in front of a line of 9 rooms. An assignment of each person to a room is called *good* if each room has at most one person using it, and no two people are using adjacent rooms. Compute the number of good assignments.

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16. [12] **Austin Wang** is making instant ramen with tofu. To make the instant ramen, Austin must perform 5 steps in the following order: add soup base, add spicy oil, add tofu, pour water, and microwave. However, Austin threw away the label, so he performed these steps in random order. Austin's microwave explodes if and only if at least one of the following conditions occur:

- The tofu is microwaved without adding water before microwaving.
- Water is poured before tofu is added and then subsequently microwaved.

Compute the probability Austin blows up his microwave.

17. [12] Let F_n denote the Fibonacci sequence, where $F_1 = 1$, $F_2 = 1$, and for $n > 2$, $F_n = F_{n-1} + F_{n-2}$. Compute

$$\sum_{n=1}^{\infty} \frac{F_n}{13^n}.$$

18. [12] Let a and b be real numbers satisfying

$$a^4 = a^2 \cdot b^2 + b^4.$$

Find the value of $(\frac{a}{b})^2$.

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19. [14] The sum of the reciprocals of the factors of 2024, $(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2024})$, can be written as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.
20. [14] The MN Youth Math Outreach board is doing a Secret Santa. **Michael Luo** and **Austin Wang** accidentally show each other the present they bought before the exchange. If there are 7 people on the board and each person brings a unique gift, how many ways are there to distribute the gifts so that no one receives a gift they've seen?
21. [14] A cone made of blue Play-Doh is covered with red Play-Doh such that the outside of the red Play-Doh forms a sphere that circumscribes the blue cone. This structure is cut into two equal hemispheres with a knife such that it passes through exactly one point on the perimeter of the base of the cone. Once cut, the flat surface on both hemispheres contains red and blue areas. If the base radius of the cone is 6, and its height is $6\sqrt{3}$, find the area of the red portion.

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22. [16] Consider a triangle ABC with lengths $AB = 13$, $BC = 14$, and $AC = 15$. Consider the circles with diameter AB and AC . Let these circles intersect at point P other than A . Let $\overline{AP}$ intersect the common external tangent of the circles at point X such that X and A are on opposite sides of $\overline{BC}$. Compute the length of AX .
23. [16] Let $a_1, \dots, a_n$ be a finite geometric sequence of positive integers with $n > 2$. Given that a_1 and a_n both have exactly 6 divisors and a_2 has exactly k divisors, find the sum of all possible values of k .
24. [16] Find the last three digits of the expression $1^{100} + 3^{100} + 5^{100} + \dots + 19^{100}$.

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25. [20] Let N be the number of primes $p < 10^6$ such that $p + 2$ is also prime. Estimate N .

An estimate of $E > 0$ earns $\lfloor 20 \min(N/E, E/N) \rfloor$ points.

26. [20] The MN Youth Math Outreach board consists of **Angie Huang, Aarya Garimella, Jinu Lee, Sneha Kundu, Ryan Niu, Michael Luo**, and **Austin Wang**, who are all from distinct high schools. Estimate the area A of the convex hull formed by these schools in square miles.

An estimate $E > 0$ earns $\lfloor 20 \min(A/E, E/A) \rfloor$ points.

(The *convex hull* of a finite set of points $\mathcal{P}$ is the smallest convex polygon which contains all points in $\mathcal{P}$. It may be visualized as a rubber band stretched around $\mathcal{P}$.)

27. [20] The MN Youth Math Outreach Board has birthdays $B_1, B_2, \dots, B_7$ when written in MMD-DYYYY form. For example, if Taylor Swift was the eighth member of the MN Youth Math Outreach Board, B_8 would be 12131989, since her birthday is December 13th, 1989. Estimate the sum S of the pairwise gcds of the B_i , or

$$S = \sum_{1 \leq i < j \leq 7} \gcd(B_i, B_j).$$

An estimate of E earns $\lfloor 20 \min(S/E, E/S) \rfloor$ points.